

PROJECT TITLE

**AI-Powered Automated Underwater Marine Debris and
Anomaly Detection System using Side-Scan Sonar Imagery**

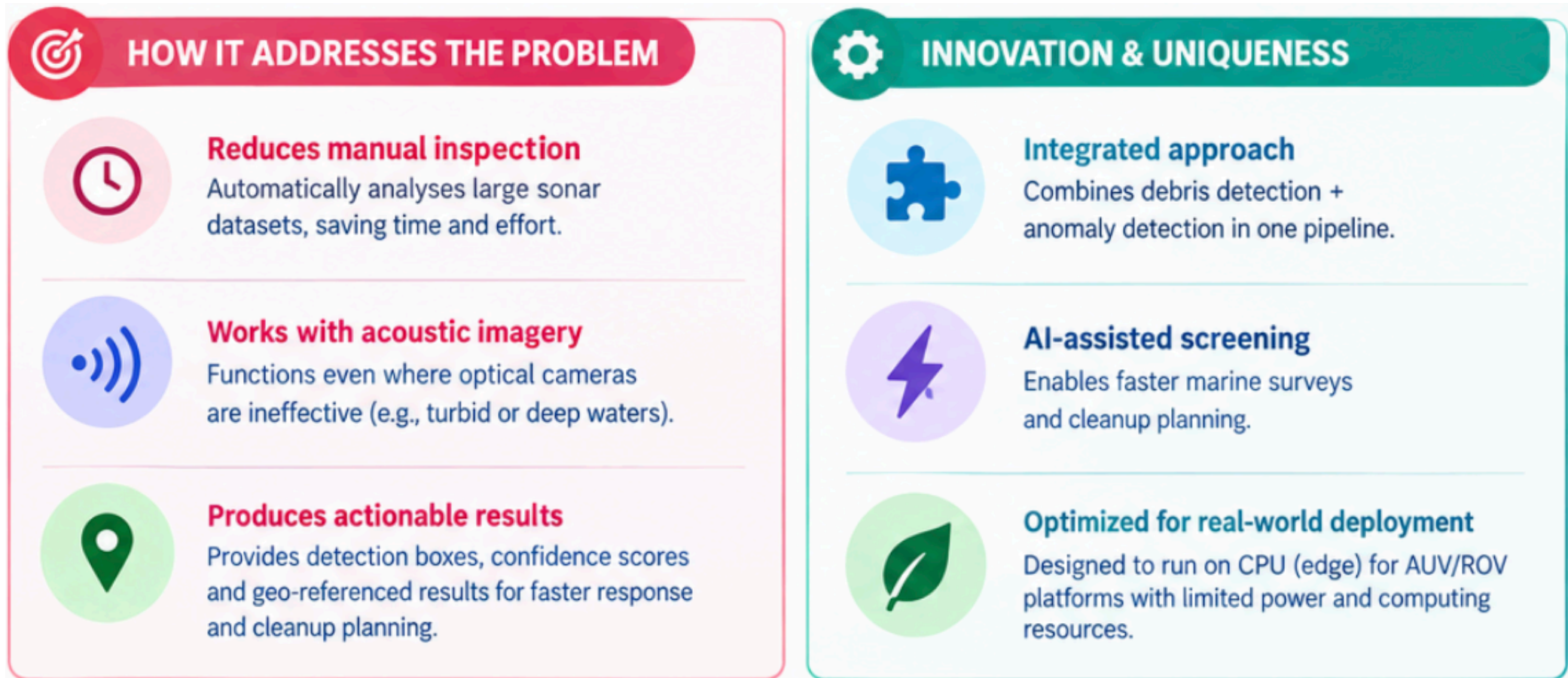
Theme: Disaster Management | Category: Software

MADE BY: Ayush kumar mishra

IDEA TITLE

PROPOSED SOLUTION

- AI system that analyses Side-Scan Sonar (SSS) imagery to detect marine debris and underwater anomalies.
- Preprocesses sonar images, detects suspicious objects, classifies them and highlights their location.



TECHNICAL APPROACH



Python

Core programming language for data processing, model development and system integration.

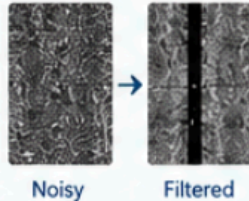
```
import cv2
import numpy as np
from flask import Flask
```



OpenCV

Image preprocessing and noise reduction.

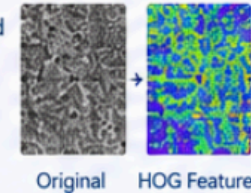
- ✓ Grayscale conversion
- ✓ Filtering (Gaussian, median)
- ✓ Contrast enhancement
- ✓ Image normalization



scikit-image

HOG/LBP feature extraction for better feature representation of sonar images.

- ✓ HOG (Histogram of Oriented Gradients)
- ✓ LBP (Local Binary Patterns)
- ✓ Feature vector generation



scikit-learn

Machine learning models for classification and anomaly detection.

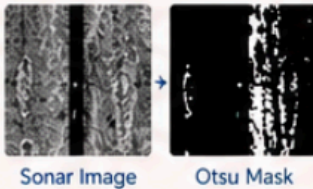
- ✓ RandomForest (debris vs background)
- ✓ IsolationForest (anomaly detection)
- ✓ Model evaluation & tuning



Classical CV Detection

Adaptive thresholding and segmentation for robust object detection.

- ✓ CFAR (Constant False Alarm Rate)
- ✓ Otsu segmentation
- ✓ Morphological operations



Flask

Web dashboard backend for data handling, model inference and API services.

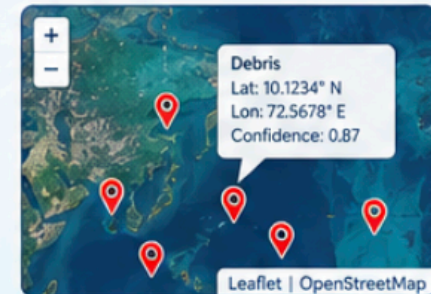
- ✓ REST APIs
- ✓ Image processing pipeline
- ✓ Real-time inference
- ✓ Result visualization



Leaflet.js + OpenStreetMap

Interactive geo-tagged map for visualizing detected debris and anomalies.

- ✓ Marker clustering
- ✓ Custom popups
- ✓ Geo-referenced coordinates
- ✓ Interactive navigation



IMPLEMENTATION FLOW



Underwater Object Detection Pipeline

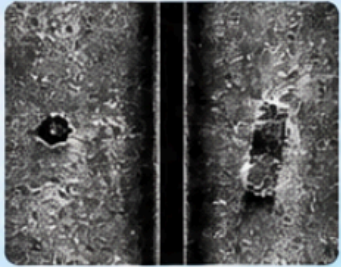
From Sonar Data to Actionable Insights

Cleaner Oceans | Safer Missions | Smarter Decisions



1

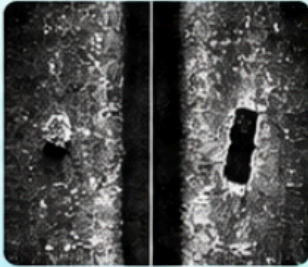
Side-Scan Sonar Data



Collect side-scan sonar data from AUV/ROV or vessel.

2

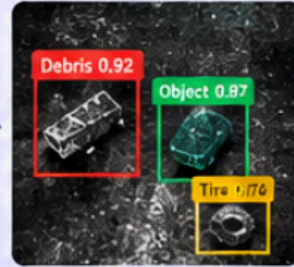
Pre-processing & Image Enhancement



Remove noise, correct distortions and enhance image quality.

3

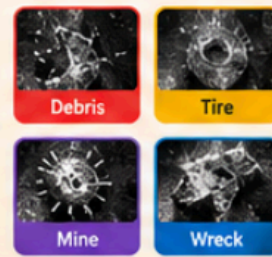
AI-Based Object Detection



Use CPU-based deep learning (no GPU) to detect potential objects in the sonar image.

4

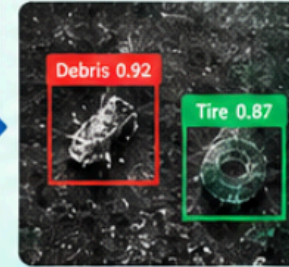
Debris / Anomaly Classification



Classify detected objects into categories (e.g., debris, mine, tire, wreck, natural feature).

5

Bounding Boxes + Confidence Scores



Mark detected objects with bounding boxes and show model confidence.

6

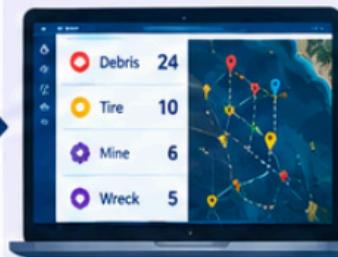
Geo-tagging & Mapping



Assign geographic coordinates and plot on a map.

7

Dashboard / Detection Report



Visualize results, filter by type, review details and generate reports for further action.

FEASIBILITY AND VIABILITY



FEASIBILITY

A practical, scalable and deployable solution for underwater debris and anomaly detection



Uses Existing Side-Scan Sonar Imagery

Leverages publicly available and commercially used side-scan sonar data as input.



CPU-only Architecture

No GPU dependency, making it genuinely deployable on AUV/ROV edge hardware.



Modular Design

Supports future AUV/ROV integration and easy scalability.



CHALLENGES & RISKS



Limited Labelled Sonar Datasets

Few publicly available datasets with debris annotations. Diverse seabed and environmental conditions are under-represented.



Low Signal-to-Noise Ratio and Sonar Artefacts

Speckle noise, shadows and reflections can lead to incorrect detections.



Similar Appearance of Debris and Natural Seabed Objects

Rocks, coral, wreckage and debris can look alike in sonar imagery.



MITIGATION



Data Augmentation + Image Enhancement

- Rotation, flip, noise addition, scaling
- Contrast enhancement & filtering



Confidence Thresholds + Human Verification

- Filter low-confidence detections
- Operator review for critical cases

IMPACT AND BENEFITS



POTENTIAL IMPACT



Faster identification of underwater debris and suspicious objects.



Supports marine pollution monitoring, survey and cleanup planning.



Helps researchers and authorities prioritise areas for inspection.



BENEFITS



ENVIRONMENTAL

Better monitoring of underwater pollution



ECONOMIC

Less manual sonar screening time



RESEARCH

Structured sonar detections for marine studies



SAFETY

Early identification of potentially hazardous objects



SCALABILITY



AUV/ROV Missions



GIS-based Dashboards



Larger Sonar Datasets

Can be scaled to AUV/ROV missions, GIS-based dashboards, and larger sonar datasets.

RESEARCH AND REFERENCES

1 NOAA Marine Debris Program

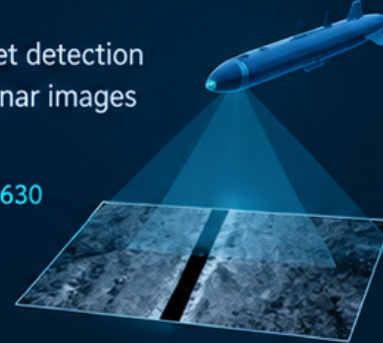
Marine debris monitoring, detection & research.



2 Li et al., Applied Ocean Research (2023)

Real-time underwater target detection for AUV using side-scan sonar images based on deep learning.

DOI: 10.1016/j.apor.2023.103630



3 Zeng et al., Ocean Engineering (2026)

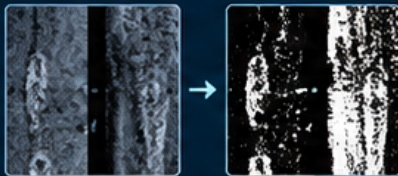
Underwater side-scan sonar object detection with multi-dimensional feature enhancement and adaptive attention routing network.

DOI: 10.1016/j.oceaneng.2026.124534



4 OpenCV

Computer vision & image processing.



5 scikit-learn

RandomForest classification and IsolationForest anomaly detection.



RandomForest



IsolationForest

6 System Connection

